

ANGCHEN XIE

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EDUCATION

Carnegie Mellon University

M.S. in Robotics (MSR)

Advised by Guanya Shi and Max Simchowitz

Pittsburgh, PA

Aug. 2025 – May 2027 (expected)

- GPA: 4.17/4.00
- Courses: Introduction to Robot Learning; Optimal Control and Reinforcement Learning; Computer Vision

Shanghai Jiao Tong University

B.Eng. in Automation

Shanghai, China

Sept. 2021 – June 2025

- GPA: 4.01/4.30 | Rank: 1/87
- Undergraduate thesis: *Fault-Tolerant Control of Legged Robots under Multi-Sensor Fusion*

RESEARCH INTERESTS

I work toward general-purpose robots that combine broad learned priors with grounded models of themselves and the physical world, enabling reliable and versatile behavior across tasks, embodiments, and environments.

PUBLICATIONS

* Equal contribution.

- [1] [Under review, CoRL 2026] Angchen Xie*, Nikhil Sobanbabu*, Ishayu Shikhare, Alan Wang, Max Simchowitz, Guanya Shi. “FADA: Few-Shot Domain Adaptation via Dynamics Alignment for Humanoid Control”.
📄 arXiv 🌐 Website 🐦 Thread
- [2] [RSS 2026] Rishabh Madan, Angchen Xie, Samantha Saak, Andres Blanco, Dohyeok Lee, Sarah Grace Brown, Yunting Yan, Mark Zolotas, Jose Barreiros, Tapomayukh Bhattacharjee. “TACTIC: Tactile and Vision Conditioned Contact-Centric Control for Whole-Arm Manipulation”.
📄 arXiv 🌐 Website
- [3] [RSS 2026] Pablo Ortega-Kral*, Eliot Xing*, Arthur Buckner, Vernon Luk, Junseo Kim, Owen Kwon, Angchen Xie, Nikhil Sobanbabu, Yifu Yuan, Megan Lee, Deepam Ameria, Bhaswanth Ayapilla, Jaycie Bussell, Guanya Shi, Jonathan Francis, Jean Oh. “RIO: Flexible Real-Time Robot I/O for Cross-Embodiment Robot Learning”.
📄 arXiv 🌐 Website 📄 Code
- [4] [CoRL 2025] Rishabh Madan, Jiawei Lin, Mahika Goel, Angchen Xie, Xiaoyu Liang, Marcus Lee, Justin Guo, Pranav N. Thakkar, Rohan Banerjee, Jose Barreiros, Kate Tsui, Tom Silver, Tapomayukh Bhattacharjee. “Prioritouch: Adapting to User Contact Preferences for Whole-Arm Physical Human-Robot Interaction”.
📄 arXiv 🌐 Website
- [5] [IROS 2024] Angchen Xie, Yeqiang Qian, Weihao Yan, Chunxiang Wang, Ming Yang. “Non-repetitive: A Promising LiDAR Scanning Pattern”.
📄 Paper 📄 Dataset

RESEARCH EXPERIENCE

LeCAR Lab, Carnegie Mellon University

Research Intern

Pittsburgh, PA

Aug. 2025 – Present

Advised by [Guanya Shi](#) and [Max Simchowitz](#)

Topic: humanoid whole-body control; infrastructure for cross-embodiment robot learning

- A Planner-IDM (inverse dynamics model) architecture that adapts to unseen terrain, payload, and actuator dynamics by finetuning only the IDM on about two minutes of target-domain rollouts, without rewards or demonstrations. (*FADA*)
- Open-source infrastructure validated on vision-language-action deployment across three morphologies and four hardware platforms. (*RIO*)

RL² Lab, Shanghai Jiao Tong University

Research Intern

Shanghai, China

Jan. 2025 – June 2025

Advised by [Yue Gao](#)

Topic: fault-tolerant locomotion for legged robots

- Model-based reinforcement learning for hexapod locomotion across varied terrain, deployed on hardware.
- A controller that adapts to the failure of one or more joints, including failure modes unseen during training.
- Multi-sensor fusion for terrain and self-failure perception, improving generalization on the real platform. (*Undergraduate thesis*)

EmPRISE Lab, Cornell University

Research Intern

Ithaca, NY

July 2024 – Dec. 2024

Advised by [Tapomayukh Bhattacharjee](#)

Topic: contact-rich whole-arm manipulation for physical human-robot interaction

- A contact-centric predictive model coupling learned latent dynamics with analytical kinematics through contact Jacobians, driven by a contact-aware sampling-based MPC planner. (*TACTIC*)
- A learning-to-rank framework that resolves conflicting user force preferences across simultaneous contacts. (*PrioriTouch*)

CyberC3 Intelligent Vehicle Lab, Shanghai Jiao Tong University

Research Intern

Shanghai, China

Nov. 2022 – June 2024

Advised by [Ye Qiang Qian](#) and [Ming Yang](#)

Topic: LiDAR perception for intelligent vehicles

- *Repetitive-or-not*, a CARLA dataset captured simultaneously by repetitive and non-repetitive LiDARs, isolating scanning pattern as the single variable.
- Showed non-repetitive scanning is the more promising pattern on the strength of its object scanning, raising the maximum detection count on small traffic participants by 59.6%, and quantified the domain gap between the two point-cloud types. (*Non-repetitive*)

SERVICE

Reviewer, Conference on Robot Learning (CoRL)

2026

Reviewer, Robotics: Science and Systems (RSS)

2026

Reviewer, IEEE Robotics and Automation Letters (RA-L)

2025, 2026

HONORS AND AWARDS

Outstanding Graduate Award, Shanghai Jiao Tong University

2025

China National Scholarship (top 0.2%)

2024

First Prize Academic Scholarship, Shanghai Jiao Tong University

2024

National First Prize, China Undergraduate Electronics Design Contest (top 1%)

2023

First Prize, China Undergraduate Mathematical Contest in Modeling (top 2%)

2022